Appendix III: Example calculation of observed statistical new distribution of unbound electrons

1. The Boltzmann Distribution in Normal Paramagnetic DPPH

In a conventional paramagnetic material like DPPH, the alignment of unbound electron spins in an external magnetic field is governed by the Boltzmann distribution. The energy difference between a spin parallel (spin up, $m_s = +\frac{1}{2}$) and antiparallel (spin down, $m_s = -\frac{1}{2}$) is

$$\Delta E = 2\mu_B B$$

where $\mu_B = 9.274 \times \frac{10^{-24}\,\mathrm{J}}{\mathrm{T}}$ is the Bohr magneton. The ratio of the number of spins parallel (N_+) to antiparallel (N_-) is

$$\frac{N_+}{N_-} = \exp\left(\frac{\Delta E}{k_B T}\right) \approx 1 + \frac{\Delta E}{k_B T},$$

with $k_B = 1.3806 \times \frac{10^{-23}\,\mathrm{J}}{\mathrm{K}}$ and room temperature $T \approx 300$ K. For a typical field $B \approx 0.1$ T,

$$\frac{\Delta E}{k_B T} = \frac{2\mu_B B}{k_B T} \approx \frac{2(9.274 \times 10^{-24})(0.1)}{(1.3806 \times 10^{-23})(300)} \approx 4.48 \times 10^{-4}.$$

Thus, the excess fraction of parallel spins is only about 0.0448%. In terms of percentages, if we consider 1,000,000 spins:

- Parallel: $\frac{1}{2}(1 + 4.48 \times 10^{-4}) \times 10^6 \approx 500{,}224$
- Antiparallel: $\frac{1}{2}(1 - 4.48 \times 10^{-4}) \times 10^6 \approx 499{,}776$

That is roughly 50.0224% parallel and 49.9776% antiparallel. The net magnetization is proportional to the tiny excess of about 448 spins out of a million, leading to the very small paramagnetic susceptibility $\chi \approx 1 \times 10^{-4}$ and relative permeability $\mu_r = 1 + \chi \approx 1.0001$.

2. What Does $\mu_r = 1.4$ Mean in Terms of Net Excess?

Our experiment yielded a measured relative permeability of $\mu_r = 1.4$ after the phase transition. This implies a magnetic susceptibility

$$\chi = \mu_r - 1 = 0.4$$

Magnetization is related to the applied field by $M = \chi H$. From our data in Table 1, for an external field $B_{\mathrm{ext}} \approx$ 4.2mT (row 1), the measured contribution of the DPPH sample was $\Delta B = 1700\mu$ T = 1.7mT. This ΔB is related to M by,

$$\Delta B = \mu_0 M \;\Rightarrow\; M = \frac{\Delta B}{\mu_0} = \frac{1.7 \times 10^{-3}\ \mathrm{T}}{4\pi \times \frac{10^{-7}\mathrm{H}}{\mathrm{m}}} \approx 1{,}353\ \mathrm{A/m}$$

Now we compare this to the theoretical saturation magnetization M_{sat} if all unbound electrons were aligned parallel. For our 36 mg DPPH sample:

- Molar mass of DPPH $\approx$ 394.32 g/mol
- Number of moles $n = \frac{0.036\text{ g}}{394.32\text{ g/mol}} \approx 9.14 \times 10^{-5}$ mol
- Number of unbound electrons $N_{\text{total}} = nN_A \approx (9.14 \times 10^{-5})(6.022 \times 10^{23}) \approx 5.50 \times 10^{19}$
- Total magnetic moment for 100% alignment: $m_{\text{total}} = N_{\text{total}}\,\mu_B \approx (5.50 \times 10^{19})(9.274 \times 10^{-24}) \approx 5.10 \times 10^{-4}\text{ A} \cdot \text{m}^2$
- Volume of sample (density $\approx$ 1 g/cm^3): $= 0.036\text{ cm}^3 = 3.6 \times 10^{-8}\text{ m}^3$
- Saturation magnetization: $M_{\text{sat}} = \frac{m_{\text{total}}}{V} = \frac{5.10\times10^{-4}}{3.6\times10^{-8}} \approx 14{,}167$ A/m.

Thus, the measured magnetization $M \approx 1{,}353$ A/m is

$$\frac{M}{M_{\text{sat}}} = \frac{1{,}353}{14{,}167} \approx 0.096 = 9.6\%$$

This $\mathbf{9.6}\%$ is the net excess alignment relative to the maximum possible excess (i.e., 100% alignment would give an excess equal to the total number of spins). In other words, out of the total N_{total} spins, the number that are excessively parallel (parallel minus antiparallel) is $0.096 \times N_{\text{total}}$.

3. Converting Net Excess to Parallel/Antiparallel Percentages

Let N_+be the number of parallel spins and $N_-$the number of antiparallel spins, with $N_+ + N_- = N_{\text{total}}$. The net excess is $N_+ - N_- = \Delta N$. We have

$$\frac{\Delta N}{N_{\text{total}}} = 0.096$$

Solving:

$$N_+ = \frac{N_{\text{total}} + \Delta N}{2} = \frac{N_{\text{total}}\,(1 + 0.096)}{2} = 0.548 N_{\text{total}},$$
$$N_- = \frac{N_{\text{total}} - \Delta N}{2} = \frac{N_{\text{total}}\,(1 - 0.096)}{2} = 0.452 N_{\text{total}}.$$

Therefore, the statistical distribution is approximately

54.8% parallel and 45.2% antiparallel alignment to external field

This is a 9.6 percentage point difference ($54.8 - 45.2 = 9.6$). In contrast, the normal paramagnetic distribution has a difference of only about 0.045 percentage points. Our experiment has increased the net alignment by a factor of roughly $9.6/0.045 \approx 213$ times, which is consistent with strongly disturbing the initial random Boltzmann distribution in our sample in favor of parallel aligned unbound electrons with the external field.

4. Summary

- Measured relative permeability: $\mu_r = 1.4$
- Implied statistical distribution of unbound electrons: approximately $\mathbf{55}$% parallel, $\mathbf{45}$% antiparallel.
- This represents a massive departure from the normal paramagnetic state (50.022%/49.978%) and is consistent with a genuine magnetic phase transition to a weakly ferromagnetic state.